



Philopateer Karam

JUNIOR HPC SYSTEM ADMINISTRATOR

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Professional Summary

Recent Computer Science graduate from Cairo University with a GPA of 3.91, specializing in High-Performance Computing, Linux systems, and cluster administration. Hands-on experience in HPC infrastructure, cluster provisioning, workload management, and parallel programming through internships and research. Skilled in Linux, MPI/OpenMP, SLURM, xCAT, and performance optimization. Currently expanding into automation and container orchestration.

Education

Faculty of Computing and Artificial Intelligence, Cairo University

2022 – 2026

Bachelor's degree in Computer Science | GPA: 3.91

Relevant Courses: High-Performance Computing & Parallel Programming, Advanced OS, Computer Architecture, Cloud Computing, Software Engineering, Data Structures and Algorithm Analysis, Computer Networks.

Technical Skills

Programming Languages: C, C++, Python, Java

High-Performance Computing: MPI, OpenMP, Parallel Programming, Performance Optimization, SLURM

Systems & Cluster Management: Linux/Unix, Bash Scripting, FreeIPA, Cluster Provisioning (xCAT)

Database Management: SQL

Version Control: Git, GitHub

Software Design: Object-Oriented Programming (OOP), Design Patterns, MVC

Technical Competencies: Performance Analysis, Debugging, Code Optimization

Soft Skills: Problem Solving, Team Collaboration

Experience

HPC Research Group Member

Cairo University | Mar 2025 – Present

Contributing to research and infrastructure projects in High-Performance Computing (HPC). Involved in cluster administration, performance optimization, and parallel computing tasks.

Tech: Linux, Bash, xCAT, FreeIPA, SLURM, C, C++, MPI, OpenMP

HPC Intern – System Support & Administration

Applied Innovation Center (AIC), MCIT | Jul 2025 – Sep 2025

Worked on “HPC System Support and Administration” project. Assisted in installation and configuration of HPC software applications, supported Linux-based cluster management, and utilized SLURM Workload Manager for job scheduling and resource allocation. Applied C, C++, and Python to support system operations and automation tasks.

Tech: Linux, Bash, xCAT, FreeIPA, SLURM, C, C++

Certificate: [AIC-Intern-Certificate](#)

HPC Intern – Cluster Infrastructure & Deployment

HPC R&D Lab, FCAI – Cairo University | Jul 2025 – Sep 2025

Contributed to rebuilding *HiPer-FC*, the HPC cluster at FCAI-CU R&D Lab, participated in setting up an experimental computer lab designed for secure and durable access. Assisted in cluster deployment, system configuration, and infrastructure support under academic supervision.

Tech: Linux, HPC Infrastructure

Certificate: [HPC-Intern-Certificate](#)

Projects

MPI-Based HPC Cluster

Distributed Systems & Parallel Computing | Mar 2025 – Jun 2025

Configured a multi-node MPI cluster and executed distributed parallel applications across compute nodes. Managed SSH trust setup and host file-based execution for efficient cluster communication.

Doc: [Beowulf Cluster Report](#)

Linux Command Line Interpreter

Systems Programming | Oct 2024

Built a C++ version of the Linux CLI from scratch, supporting commands like `cp`, `pwd`, `ls`, `rm`, `mkdir`, `mv`, pipes (`|`), and redirection (`>>`, `<<`).

Repo: [Linux-Command-Line-Interpreter](#)

CPU Scheduler Simulator

Operating Systems & GUI | Nov 2024

Designed and implemented a GUI-based CPU scheduling simulator supporting four algorithms: Shortest Remaining Time First, Shortest Job First, Priority Scheduling, and the custom FCAI Scheduler.

Repo: [CPU-Scheduler-Simulator](#)

AI Vision Classifier System

Python, Machine Learning, Computer Vision | Dec 2025

Developed an end-to-end computer vision system for real-time material classification using deep feature extraction and classical machine learning classifiers. The system was designed to be robust against lighting variations, motion blur, and sensor noise.

Repo: [AI-Vision-Classifier-System](#)

Learning Management System (LMS) API

Back-End Development | Java, Spring Boot | Dec 2024

Developed a RESTful API for a web-based LMS to manage users, courses, assignments, quizzes, attendance, and notifications, enabling seamless interaction within an educational environment.

Repo: [Web-Based-Learning-Management-System](#)

Courses & Certifications

Red Hat System Administration I (RH124) | Red Hat Academy | *Jun 2026* -- [Certificate](#)

Red Hat System Administration II (RH134) | Red Hat Academy | *Jun 2026* -- [Certificate](#)

High-Performance and Parallel Computing Specialization | University of Colorado Boulder | Coursera | *Mar 2026* -- [Certificate](#)

- Introduction to High-Performance and Parallel Computing -- [Certificate](#)
- Efficient Programming -- [Certificate](#)
- Parallel Computing with MPI -- [Certificate](#)

Ubuntu Linux Essentials | MaharaTech | *Feb 2026* -- [Certificate](#)

Introduction to HPC | MaharaTech | *Jun 2025* -- [Certificate](#)

Professional Development (In Progress)

- Red Hat Enterprise Linux Automation with Ansible -- Red Hat Academy
- Introduction to Kubernetes -- The Linux Foundation